

Class Activity: 01

Scenario 1: Student Academic Performance Dashboard

Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_excel("Student Performance.xlsx")

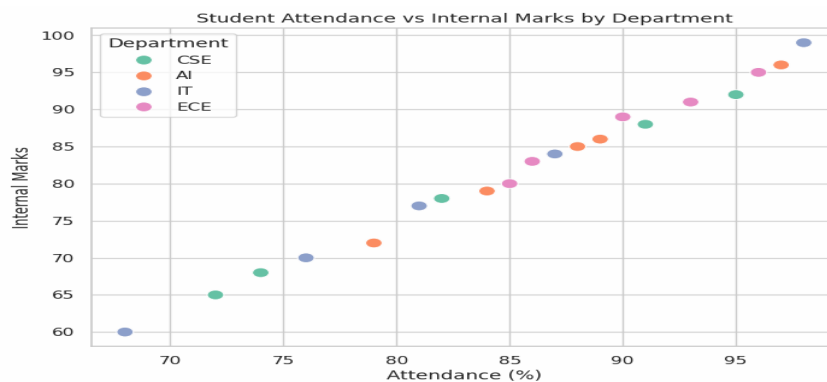
sns.set(style="whitegrid")

plt.figure(figsize=(8,6))
sns.scatterplot(
    data=df,
    x="Attendance",
    y="Internal Marks",
    hue="Department",
    palette="Set2",
    s=100
)

plt.title("Student Attendance vs Internal Marks by Department")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")
plt.legend(title="Department")

plt.show()
```

Output:



Scenario 2: Electric Vehicle Market Analysis

Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

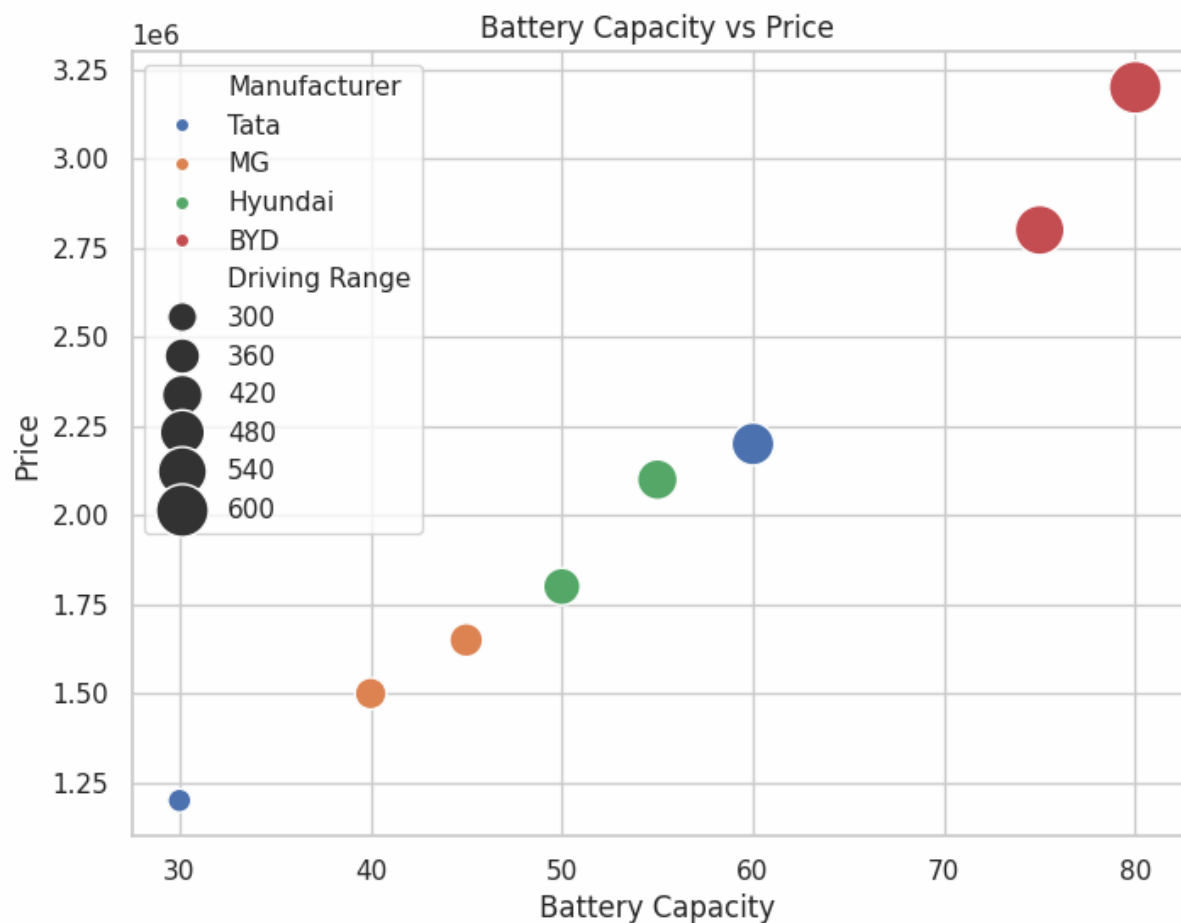
df = pd.read_excel("Electric Vehicle Market.xlsx")

plt.figure(figsize=(8,6))

sns.scatterplot(data=df,
                x="Battery Capacity",
                y="Price",
                size="Driving Range",
                hue="Manufacturer",
                sizes=(100,500))

plt.title("Battery Capacity vs Price")
plt.show()
```

Output:



Scenario 3: Hospital Patient Admission Analysis

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("Patient Admission.xlsx")

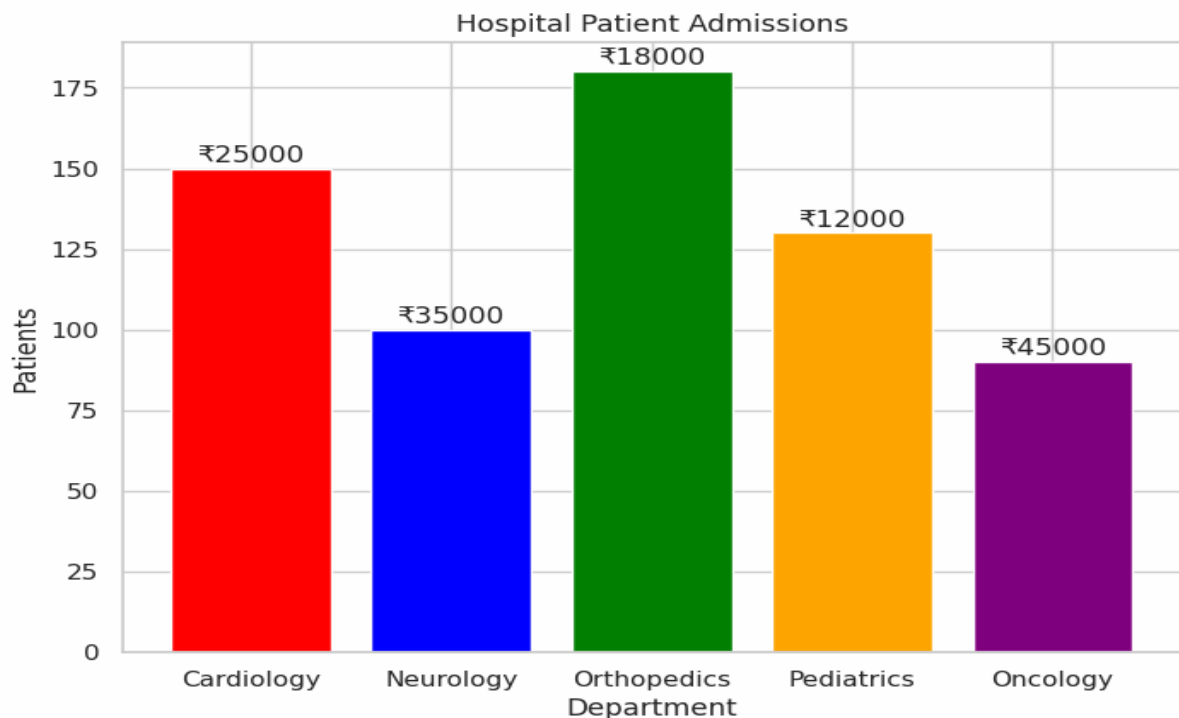
plt.figure(figsize=(8,6))

bars = plt.bar(df["Department"],
               df["Number of Patients"],
               color=["red","blue","green","orange","purple"])

for bar,cost in zip(bars,df["Average Treatment Cost"]):
    plt.text(bar.get_x()+bar.get_width()/2,
             bar.get_height()+2,
             f"₹{cost}",
             ha="center")

plt.title("Hospital Patient Admissions")
plt.xlabel("Department")
plt.ylabel("Patients")
plt.show()
```

Output:



Scenario 4: Weather Monitoring System

Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

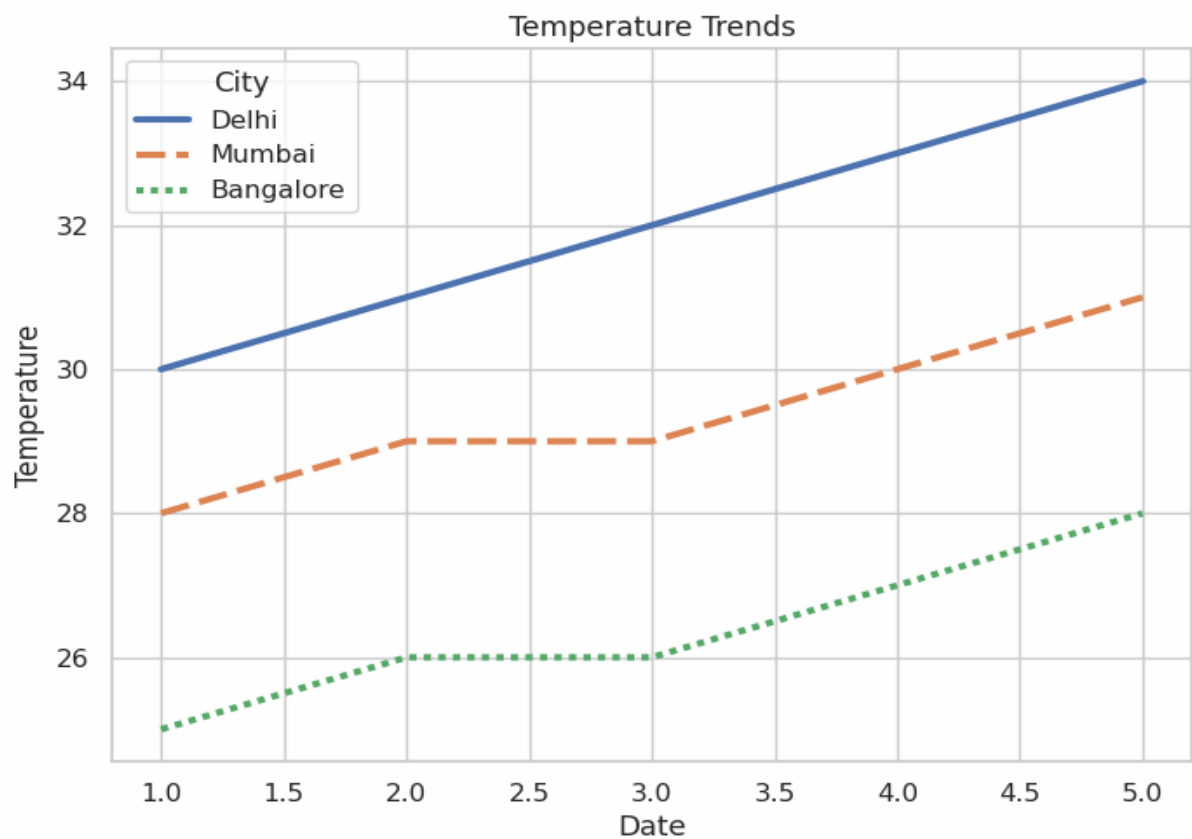
df = pd.read_excel("Weather.xlsx")

plt.figure(figsize=(8,6))

sns.lineplot(data=df,
             x="Date",
             y="Temperature",
             hue="City",
             style="City",
             linewidth=3)

plt.title("Temperature Trends")
plt.show()
```

Output:



Scenario 5: Supermarket Sales Analysis

Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

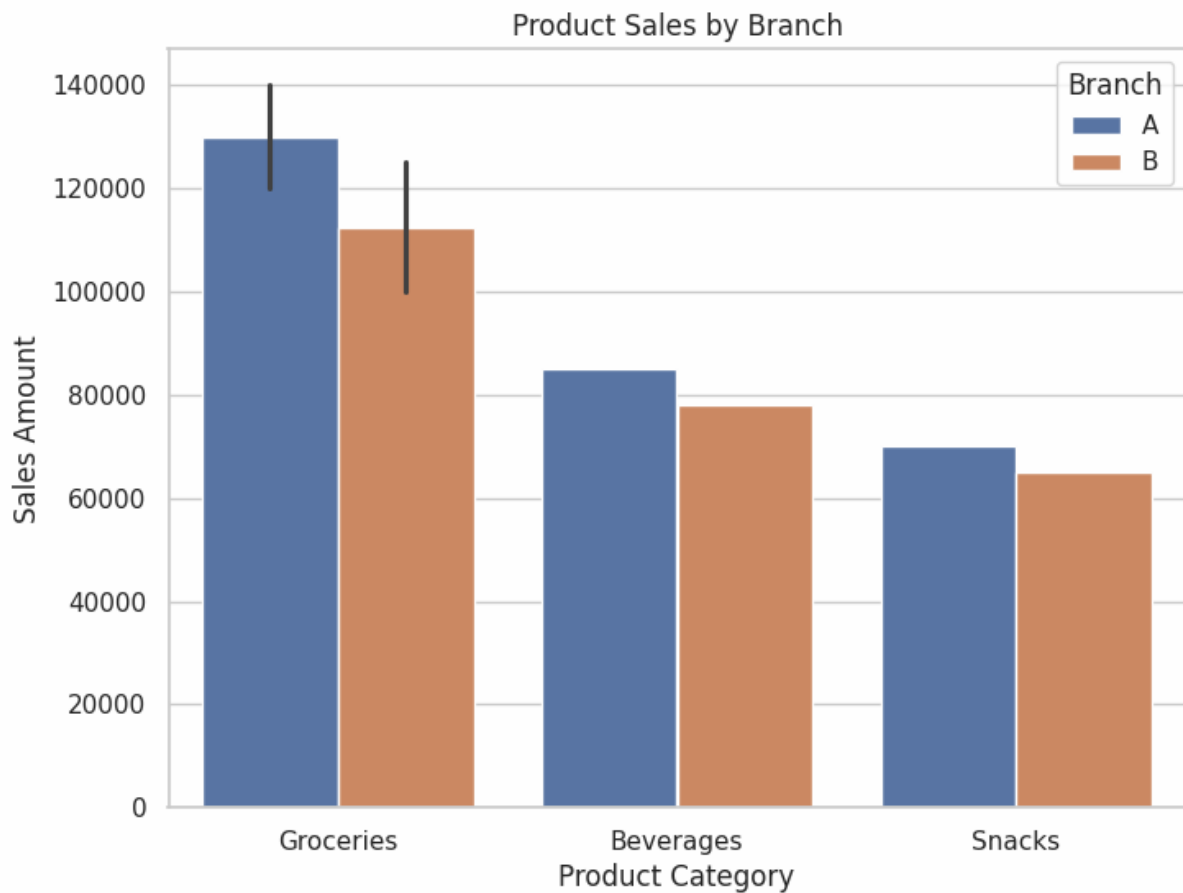
df = pd.read_excel("SuperMarketSales.xlsx")

plt.figure(figsize=(8,6))

sns.barplot(data=df,
            x="Product Category",
            y="Sales Amount",
            hue="Branch")

plt.title("Product Sales by Branch")
plt.show()
```

Output:



Scenario 6: Employee Salary Distribution

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("EmployeeSalary.xlsx")

plt.figure(figsize=(8,6))

plt.hist(df["Salary"],
        bins=5,
        color="skyblue",
        edgecolor="black")

plt.title("Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Employees")

plt.show()
```

Output:



Scenario 7: Smartphone Market Comparison

Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

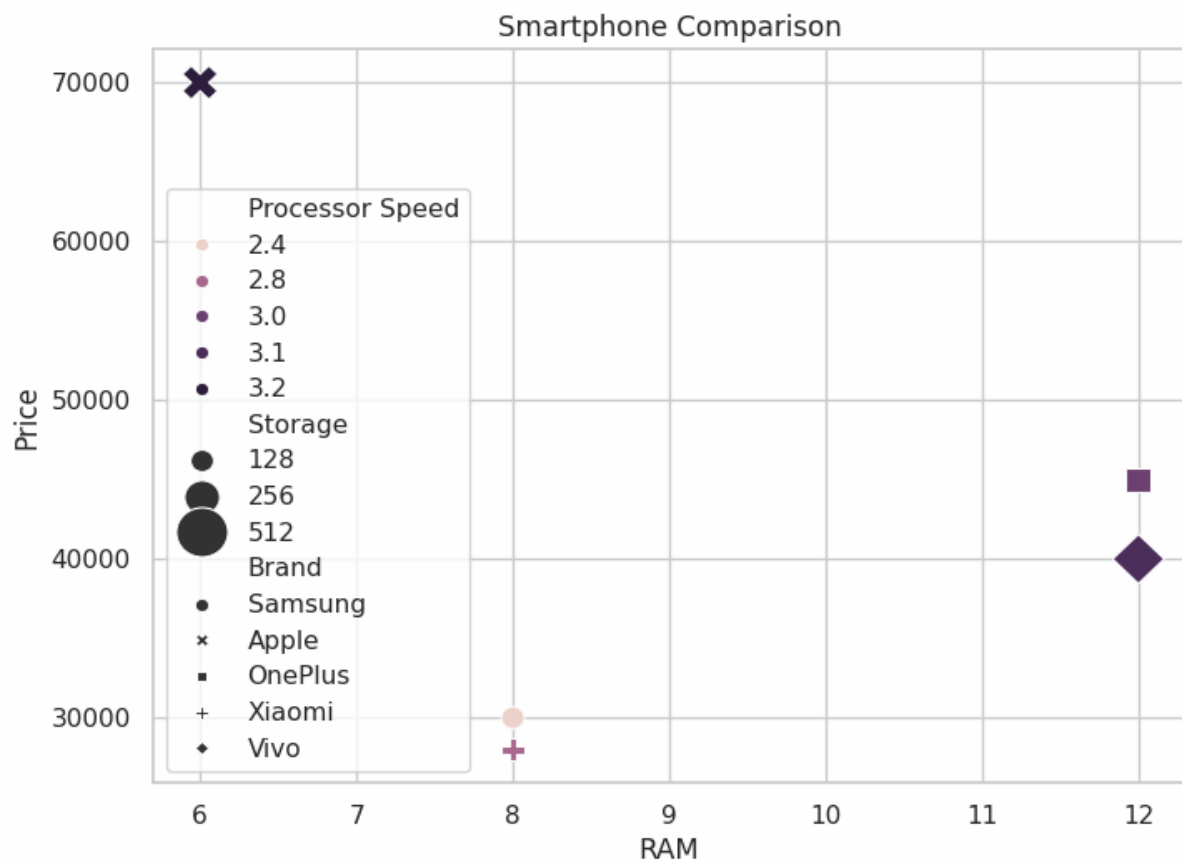
df = pd.read_excel("SmartphoneMaarket.xlsx")

plt.figure(figsize=(8,6))

sns.scatterplot(data=df,
                x="RAM",
                y="Price",
                hue="Processor Speed",
                style="Brand",
                size="Storage",
                sizes=(100,500))

plt.title("Smartphone Comparison")
plt.show()
```

Output:



Scenario 8: Online Movie Rating Analysis

Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

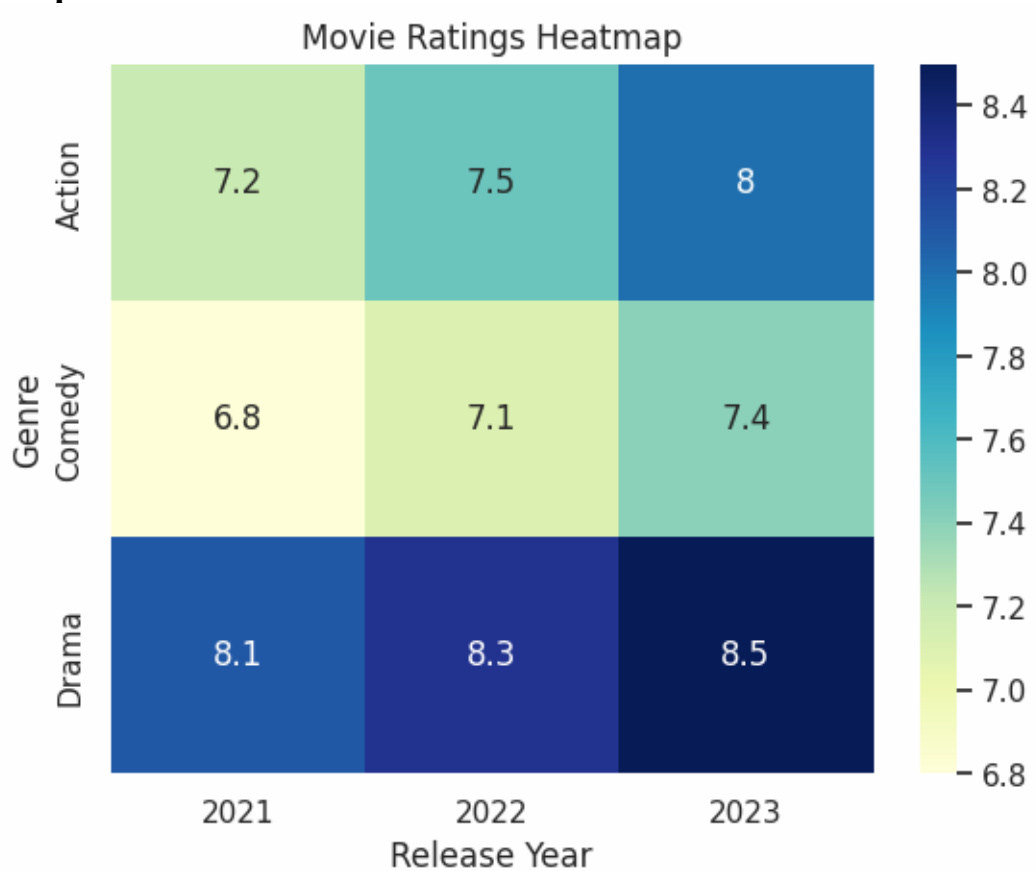
df = pd.read_excel("OnlineMovie.xlsx")

pivot = df.pivot(index="Genre",
                  columns="Release Year",
                  values="Average Rating")

sns.heatmap(pivot,
            annot=True,
            cmap="YlGnBu")

plt.title("Movie Ratings Heatmap")
plt.show()
```

Output:



Scenario 9: COVID-19 State-wise Analysis

Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

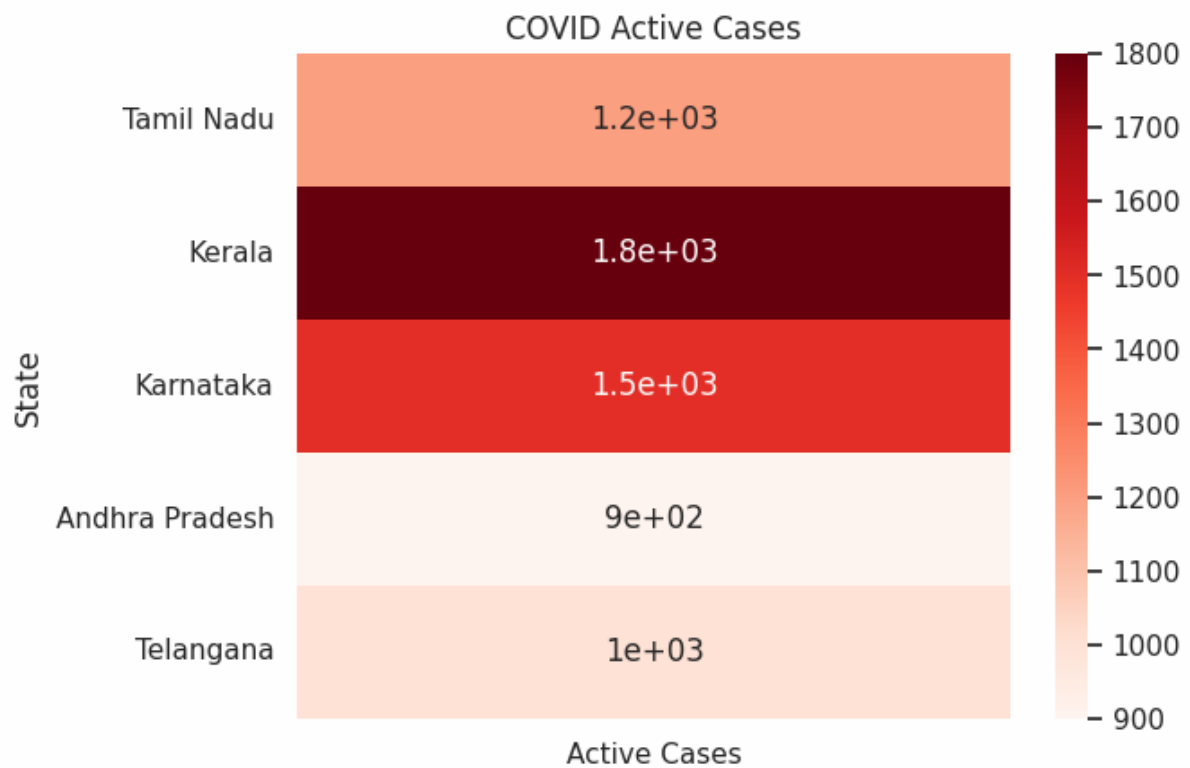
df = pd.read_excel("Covid-19.xlsx")

data = df.set_index("State")

sns.heatmap(data,
            annot=True,
            cmap="Reds")

plt.title("COVID Active Cases")
plt.show()
```

Output:



Scenario 10: Iris Flower Classification

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
iris = sns.load_dataset("iris")
```

```
plt.figure(figsize=(8,6))
```

```
sns.scatterplot(data=iris,
                x="sepal_length",
                y="petal_length",
                hue="species",
                style="species",
                size="sepal_width",
                sizes=(50,300))
```

```
plt.title("Iris Flower Classification")
```

```
plt.show()
```

Output:

